

Characteristics of runners meeting acute kidney injury criteria following a 161-km ultramarathon

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The incidence of acute kidney injury (AKI) associated with ultramarathon running is varied partially because onsite measurement of serum creatinine concentration (Cr) is rarely performed at races due to logistical challenges. We conducted a prospective observational study using blood samples from volunteer athletes before and immediately after the Leadville Trail 100 Run in August 2014. The goals of our study were to determine the incidence of those meeting the AKI criteria immediately following the ultramarathon race and to evaluate the utility of using medical equations to back-calculate creatinine levels for runners without known baseline creatinine levels and to identify possible risk factors for developing AKI. We were able to collect pre-race and post-race blood samples on 64 and 84 volunteers, respectively. AKI was defined as a Cr increase (≥ 1.5 times or ≥ 0.3 mg/dl from baseline). Of the 37 runners for whom we had both pre- and post-race blood samples, 18 (49%) met the criteria for AKI. Using two different methods of back-calculating baseline estimated creatinine (a fixed eGFR [estimated glomerular filtration rate] and an aged-based eGFR), 6 (16%) and 22 (59%) of 37 runners met the criteria for AKI, respectively. We found that there is no accurate way to estimate baseline Cr in this population.

KEYWORDS

creatinine, foot race, mountain

1 | INTRODUCTION

Ultramarathons are foot races exceeding the official marathon distance (42.195 km) and generally occur on trails that traverse remote areas.^{1,2} The popularity of ultramarathon races has exponentially grown over the past decades.¹⁻³ Due to the inherent physical stress of the races coupled with the increase in participation, the potential for serious adverse health events is growing.

Our understanding of the physiologic changes resulting from these physically demanding ultramarathons is improving. Acute kidney injury (AKI) is a clinical condition characterized by a rapid and often reversible reduction in kidney function which is usually measured by glomerular

filtration rate (GFR). As it may take time to see increases in blood urea nitrogen (BUN) or serum creatinine concentration (Cr), decline in urine output may be the only sign of AKI in the early stages which is usually non-specific and clinically not very helpful.^{4,5} In order to detect changes in GFR, it is essential to have or to estimate baseline GFR. Most equations for estimating GFR are based on the general population and hospitalized individuals.⁶⁻⁹ There is a lack of data on the normal GFR range among healthy ultramarathon runners.¹⁰ AKI is relatively common in ultramarathon competitors, with documented incidence being 25%-80%.¹⁰⁻¹⁴ Fluid and electrolyte disturbances, hemolysis and skeletal muscle damage can occur during long-distance running and a combination of dehydration, heat

stress, muscle injury, and non-steroidal anti-inflammatory drugs (NSAIDs) use may all contribute to the development of AKI.¹⁵⁻¹⁸ In general, ultramarathon runners recover from ultramarathons without medical intervention; however, development of AKI requiring hospitalization does occur.^{13,19-21}

Although AKI requiring intervention among ultramarathon participants is rare, the identification of those with AKI is valuable for medical providers so the appropriate medical interventions can be initiated.²² Potential impending renal failure is a critical diagnosis that could be avoided with prompt diagnosis and appropriate management of AKI.

The data collected for the present paper were part of a wider study with three aims: 1) determine whether post-race urine dipstick testing is useful in the early post-race identification of athletes at risk for AKI following completion of a 100-mile ultramarathon²³; 2) investigate any possible correlation between athlete demographic characteristics (eg, acclimatization pattern) and risk for AKI; and 3) identify any possible correlation between race time and variations in urine dipstick and metabolic biomarkers.²³⁻²⁵ The aim of this particular paper was a part of aim 2. The goals were to determine the incidence of those meeting the AKI criteria in this race and to evaluate the use of medical equations to back-calculate creatinine levels for runners without known baseline creatinine levels and to identify possible risk factors for developing AKI. We hypothesized that runners with abnormal urine dipstick parameters would be more likely to meet the AKI criteria than those with normal urine dipsticks.

2 | METHODS

The study was conducted in August 2014 during the Leadville Trail 100 Run near Leadville, Colorado. The 161-km ultramarathon course traverses mainly forest trails with some mountain roads and ranges in altitude from 2800 m to 3840 m.²⁴⁻²⁶ Meteorological conditions during the race were determined using data from nearby weather stations (<https://www.ncdc.noaa.gov/>). Race temperature at the start (4:00 AM) and overnight was 2°C with a light wind. The temperature rose to about 22°C and the wind reached about 16 km/h by late afternoon. The race day was dry with no precipitation and average humidity of 57% (range 22%-92%). In order to successfully complete the race, runners had to pass the finish line before 10:00 AM on the next day (30-hour cut-off).

All runners who were interested in the study were able to participate. There were no exclusion criteria. Participants were recruited during the pre-race medical check-in and at the post-race finish line. Consent was obtained prior to participation. We did not offer any compensation for participating in the study, but the athletes were informed about their test

results. The Colorado Multiple Institutional Review Board approved the study. We collected the blood and urine samples the day before the race and at the finish line whether the runners finished the race or dropped out of the race. A total of 3 mL of blood were drawn while seated into heparinized tubes through an antecubital vein. Blood samples were immediately refrigerated and then transported to the University of Colorado Hospital laboratory within 24 hours. Each sample was analyzed for comprehensive metabolic panel (blood glucose, electrolytes, kidney, and liver functions), lactate dehydrogenase (LDH), and serum creatine kinase (CK).²⁵

AKI is defined as a Cr increase (≥ 1.5 times or ≥ 0.3 mg/dL from baseline) according to the Kidney Disease Improving Global Outcomes (KDIGO) Group.²⁷ AKI has also been defined based on urine output and decreased glomerular filtration rate (GFR).²⁷⁻²⁹ As it is impractical to obtain baseline blood samples on all runners before a race, there are many different ways that have been used to estimate the baseline Cr or GFR. Most studies on ultramarathon runners have used an age-based estimation of GFR (100 mL/min for individuals ≤ 40 years of age and 140-age for individuals >40 years of age).^{10,11,19,30} Then the baseline Cr was back-calculated by rearrangement of the modified diet in renal disease (MDRD) equation:

$$\text{eGFR} = 175 \times (\text{SCr})^{-1.154} \times (\text{age})^{-0.203} \times 0.742 \text{ [if female]} \times 1.212 \text{ [if Black]}.$$

where eGFR (estimated glomerular filtration rate) is in units of mL/min/1.73 m², SCr (standardized serum creatinine) is in units of mg/dL, and age is in years.

Another way to calculate eGFR is with the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formula:

$$\text{eGFR} = 141 \times \min(\text{SCr}/\kappa, 1)^{\alpha} \times \max(\text{SCr}/\kappa, 1)^{-1.209} \times 0.993^{\text{Age}} \times 1.018 \text{ [if female]} \times 1.159 \text{ [if Black]}.$$

where κ is 0.7 for females and 0.9 for males, α is -0.329 for females and -0.411 for males, min indicates the minimum of SCr/ κ or 1, and max indicates the maximum of SCr/ κ or 1.

The KDIGO Group recommends using a baseline eGFR of 75 mL/min/1.73 m² for those without a baseline Cr.²⁷ The National Kidney Foundation (NKF) recommends using the CKD-EPI or MDRD equations for eGFR (<https://www.kidney.org>). The NKF uses a GFR of 45 to 59 mL/min/1.73 m² as mild to moderate loss of kidney function.

We used a baseline eGFR of 75 mL/min/1.73 m² and an aged-based eGFR to back-calculate the baseline Cr using the MDRD equation on runners whom baseline Cr was not measured. We also used the MDRD and CKD-EPI equations to calculate the eGFR of all runners with post-race Cr values.

Runners' characteristics, medical history, and race performance were available through online pre-race and post-race questionnaires and other studies associated with the race. Information on methodology and other analyses of this study is available elsewhere.^{24,25} Results from urine analysis are presented in another paper.²³

TABLE 1 Demographic information of participants

Participants' characteristics	Subjects with pre-race blood samples (n = 64)	Subjects with post-race blood samples (n = 84)	Subjects with both pre- and post-race blood samples (n = 37)	All runners in the race (n = 669)
Age (y); mean $\pm$ SD*	43 $\pm$ 10	42 $\pm$ 9	43 $\pm$ 10	40 $\pm$ 9
Female sex; n (%)	13 (20)	15 (18)	8 (22)	109 (16)
BMI [†] (kg/m ²); mean $\pm$ SD*	23.3 $\pm$ 2.4	22.8 $\pm$ 2.1	22.7 $\pm$ 2.4	23.4 $\pm$ 2.3
Successfully completed the race; n (%)	30 (46.9)	66 (78.6)	28 (75.7)	352 (52.6)

*Standard deviation; [†]body mass index

2.1 | Statistical analysis

We conducted statistical analyses with the IBM SPSS, version 26.0 (SPSS, Inc, Chicago, Illinois). We used descriptive statistics to examine the frequency, mean, and range for all variables (where appropriate). A Wilcoxon matched-pair test for paired samples was used to assess the impact of the race on the data obtained in the two pre-determined moments (pre- and post-race). The *P*-values $< .05$ were considered statistically significant. As Cr values were normally distributed (one-sample Kolmogorov-Smirnov test was $P = .197$), the paired samples *t* test was used to evaluate variables from individuals with pre- and post-race samples. The independent-samples *t* test was used to compare all pre-race and post-race samples. The non-parametric (Wilcoxon and Mann-Whitney *U*) analysis was used for categorical variables. Variables with significance level less than 0.05 in univariate analysis were considered in the multiple regression analysis. A stepwise procedure called forward selection was applied to include all the eligible variables based on the Akaike Information Criteria (AIC).

3 | RESULTS

Out of 669 athletes who started the ultramarathon, 352 (52%) successfully completed the race in less than 30 hours (Table 1). Women were the minority (16%) and the average age was 40 years. A total of 111 runners volunteered to participate in the study. Pre-race blood samples were collected from 64 runners, and 84 provided post-race samples. We were able to collect both pre- and post-race blood samples from 37 runners (Table 1).

The mean ($\pm$ SD) pre-race Cr among volunteers before the race ($n = 64$) was 0.99 ± 0.15 mg/dL which was lower than the estimated (back-calculated) baseline value of 1.09 ± 0.11 mg/dL using a fix eGFR of 75 mL/min/1.73 m² ($P < .001$), and higher than the value of 0.85 ± 0.09 mg/dL using an age-based eGFR ($P < .001$) (Figure 1). Thus,

back-calculating Cr value using a fix eGFR of 75 mL/min/1.73 m² resulted in 10% overestimation of the actual Cr value. On the other hand, back-calculating Cr value using an age-based eGFR resulted in 14% underestimation of the actual Cr value.

Serum BUN and Cr increased ($P < .001$) after the race among the 37 runners with both pre- and post-race samples (Figure 2). This pattern exists for the total number of runners with pre-race ($n = 64$) and with post-race ($n = 84$) samples (Figure 2). From 37 runners with both pre- and post-race blood samples, 18 (49%) met the criteria for AKI (Figure 3). Using a fixed eGFR of 75 mL/min per 1.73 m² and an age-based eGFR, 6 (16%) and 22 (59%) of these 37 runners met the criteria for AKI, respectively.

When back-calculating baseline Cr of all runners with post-race samples ($n = 84$), 18 (21%) met the criteria for AKI based on fixed eGFR assumption, and 53 (63%) met the criteria for AKI based on an age-based eGFR assumption. The mean ($\pm$ SD) eGFR among runners with post-race samples ($n = 84$) was 73 ± 18 mL/min/1.73 m² using the CKD-EPI equation and 66 ± 15 mL/min/1.73 m² using the MDRD equation ($P < .001$). Out of 84 runners with post-race samples, using the CKD-EPI, 19 (23%) had eGFR of < 60 mL/min/1.73 m² (at least mild to moderate loss of kidney function). Using the MDRD equation, 32 (38%) of runners had eGFR of < 60 mL/min/1.73 m².

Table 2 summarizes the correlations of individual characteristics with AKI criteria being met on 37 runners for which we had both pre- and post-race blood samples. In the univariate logistic regression, AKI criteria being met was associated with sex, race finish time (among finishers), and percentage of weight change. The binary multiple logistic regression showed only race finish time (among finishers) and percentage of weight change as significantly correlated with AKI criteria being met (Table 2). There were no significant relationships for months of training for the race, history of previous Leadville Trail 100 Run completion, and any other ≥ 100 -mile ultramarathon race completion with whether or not AKI criteria was met.

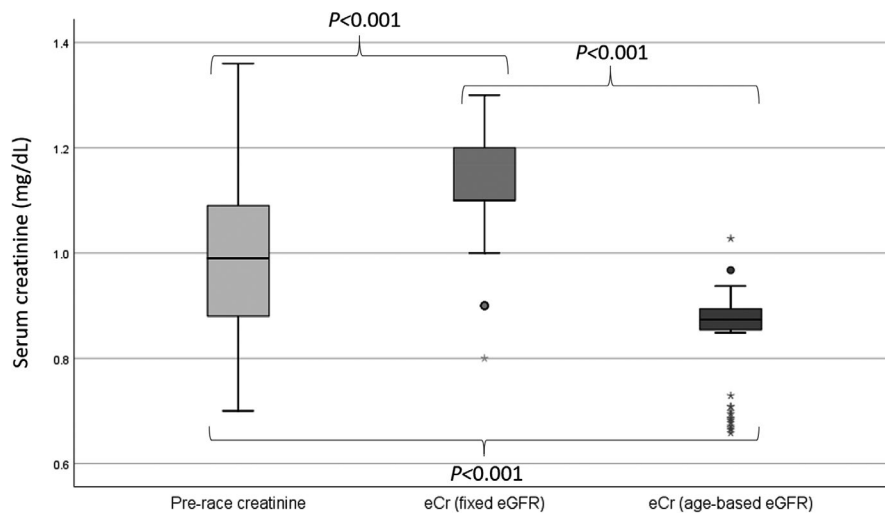


FIGURE 1 Box plot of the actual pre-race serum creatinine concentration (Cr) versus estimated (back-calculated using MDRD) baseline creatinine (eCr). This was based on eGFR assumption of 75 mL/min per 1.73 m² (fixed eGFR) and based on age-based eGFR (age-based eGFR) among 64 runners. Boxes represent 25th to 75th percentile, solid lines in the middle of the box represent the median, whiskers represent extreme values, and stars/circles represent outliers

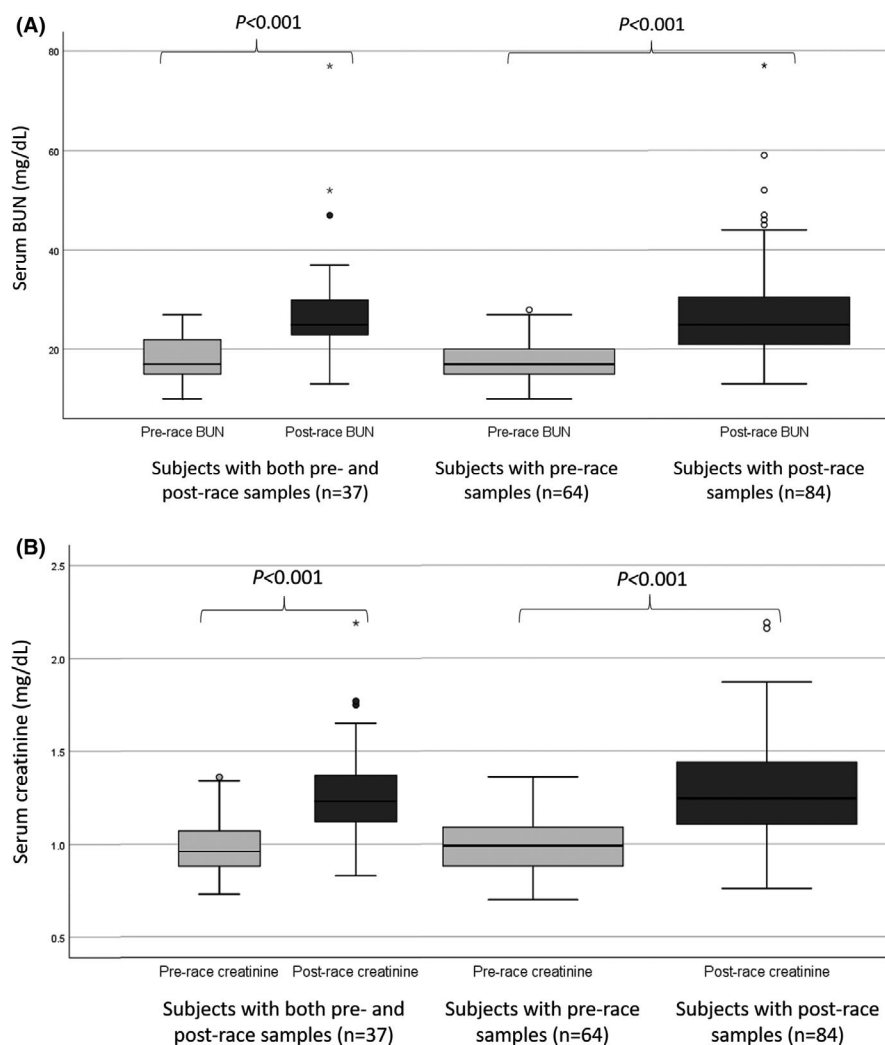


FIGURE 2 Box plot of the serum BUN (A) and creatinine (B) among runners with both pre- and post-race samples, runners with pre-race samples, and runners with post-race samples. Boxes represent 25th to 75th percentile, solid lines in the middle of the box represent the median, whiskers represent extreme values, and stars/circles represent outliers

4 | DISCUSSION

This work demonstrates that, depending on the methodology used, AKI criteria was met by 21%-63% of the subjects following the ultramarathon. This is consistent with

prior work demonstrating a 25%-80% incidence of the criteria being met during ultramarathons of various distances.^{11,14,30,31} The implications of such high incidents of AKI criteria being met are not clear, but the current belief is that there is no reason for concern about long-term

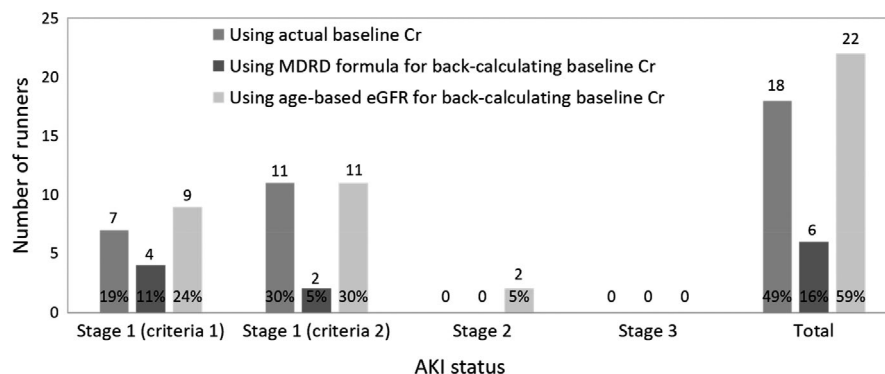


FIGURE 3 AKI status of runners with both pre- and post-race blood samples ($n = 37$). First column represents all runners with actual baseline Cr. The second column represents all runners with back-calculated baseline creatinine using MDRD formula based on a fixed eGFR of 75 mL/min per 1.73 m². The third column represents all the runners with back-calculated baseline creatinine using age-based eGFR.

harm to the kidneys in ultramarathon runners.^{31,32} It would certainly be helpful, however, to have longer-term data to evaluate potential risk in ultramarathon runners over a greater time period and larger numbers of ultramarathon races.

This work demonstrates a basic methodological issue with estimation of baseline eCr (estimated Cr) in healthy runners for use in determining post-race AKI. This problem has been previously commented upon.^{7,10,11,31-33} We used two different methods, which showed overestimation and underestimation of the number of runners meeting AKI criteria. Interestingly, using a fixed eGFR to back-calculate the eCr yielded a higher value than the measured value. This led to a large underestimation of those meeting the AKI criteria. When we used an age-based eGFR and then back-calculated the eCr, we found similar results that have been demonstrated by others, where there are large disparities and overinflated rates of AKI.¹⁰

Although, we had a relatively small number ($n = 37$) of runners with both pre- and post-race samples, this may not truly reflect the detected Cr changes across the larger sample when considering only post-race values ($n = 84$). The proportion of runners that did meet AKI criteria in this sample group (47%) is consistent with prior studies, per above (25%-80%). Therefore, the more likely issue is that using the eCr calculation with either fix eGFR assumption or age-based eGFR are simply inaccurate in this population. Given that the MDRD equation used to measure eCr is based only on age, sex, and race, it does not account for several factors that could be important in determining baseline Cr in this population. One such important influence on Cr in athletes is BMI.³⁴ As a large proportion of ultramarathoners have low BMI values,³⁵ this could certainly affect Cr. Level of fitness and other physiologic factors may also contribute but are more difficult to measure. Better calculations of eCr that consider some of these factors should be explored for use in this and other athletic populations.

Those meeting the AKI criteria were found to have run the race faster, lost more bodyweight and had higher post-race CK concentrations. High CK concentrations is

a generally known risk for development of AKI,³⁶ greater weight loss,^{11,30} and a faster race time¹⁹ have been suggested as possible risk factors. Of importance were the findings from the present work demonstrating no evidence of an association between meeting the AKI criteria and the use of NSAIDs before or during an ultramarathon. Use of NSAIDs has generated considerable concern since these drugs can reduce renal blood perfusion.³⁷ But support for this concern has been variable with some studies suggesting a risk¹⁶ and others not.^{11,38} The present findings offer no evidence for concern that NSAIDs increase the risk of acute kidney injury in ultramarathons, but the long-term effects remain unknown.

An obvious limitation in the study is the relatively small number of participants that provided both pre- and post-race blood samples (37 runners). Fortunately, we did have a larger number of participants who provided only a pre-race sample (64 runners) or only a post-race sample (84 runners), which allowed us to analyze a larger amount of data. Notable was the finding that results were similar whether comparing paired or unpaired samples. As we may not have a large sample size, some of the statistical differences seems to be small and sometimes not clinically relevant. Additionally, our sample population who gave both pre- and post-race blood samples included only 8 females (21%). While this is consistent with female participation rates in ultramarathons,³⁵ it is likely too small of a sample size from which we can draw definitive conclusions about AKI risk related to sex. Another limitation of our study with multiple data points and measurement²³⁻²⁵ is the risk of false positive findings and type 1 error.³⁹

5 | CONCLUSIONS

We found about half of the runners with pre- and post-race samples met the criteria for AKI. In this study, we found out that there is no accurate way to estimate baseline Cr in this

TABLE 2 Odds ratio estimates predicting AKI criteria being met (n = 18) among 37 runners with both pre- and post-race blood samples

	Subjects meeting AKI criteria (n = 18)	Subjects not meeting AKI criteria (n = 19)	Odds Ratio (95% Confidence Interval)	Univariate logistic regression model; <i>P</i> value	Odds Ratio (95% Confidence Interval)	Binary multiple logistic regression model; <i>P</i> value
Age (year); mean $\pm$ SD	41 $\pm$ 10	45 $\pm$ 10	0.95 (0.89-1.02)	.221		
Sex; n (%)						
Male	17 (58.6)	12 (41.4)	9.91 (1.07-91.5)	.043*	245.5 (0.55-1095)	.077
Female (RL)	1 (12.5)	7 (87.5)	-	-	-	-
BMI (Kg/m ²); mean $\pm$ SD	23.4 $\pm$ 2.4	22.1 $\pm$ 2.2	1.28 (0.93-1.76)	.125		
Race completion status; n (%)						
Finished	15 (53.6)	13 (46.4)	2.30 (0.47- 11.1)	.297		
Did not finish (RL)	3 (33.3)	6 (66.7)	-	-		
Race finish time [¥] (hours); mean $\pm$ SD	26.0 $\pm$ 3.1	28.1 $\pm$ 1.6	0.70 (0.49-0.99)	.045*	0.34 (0.12-0.91)	.033*
Percentage of weight change [¥] $\pm$ SD	-1.3 $\pm$ 1.9	0.3 $\pm$ 1.7	0.57 (0.33-0.97)	.042*	0.19 (0.05-0.76)	.019*
Use of NSAIDs 1-2 d before the race						
Yes	3 (42.9)	4 (57.1)	0.75 (0.14-3.94)	.734		
No (RL)	15 (50.0)	15 (50.0)	-	-		
Use of NSAIDs during the race						
Yes	4 (50.0)	4 (50.0)	1.07 (0.22-5.12)	.931		
No (RL)	14 (48.3)	15 (51.7)	-	-		
Post-race CK concentration (U/L); mean $\pm$ SD	19,306 $\pm$ 17,748	10,331 $\pm$ 10,081	1.00 (1.00-1.01)	.098		
Post-race proteinuria						
Yes	10 (62.5)	6 (37.5)	2.77 (0.66-11.6)	.162		
No (RL)	6 (37.5)	10 (62.5)	-	-		
Post-race urine specific gravity [†] ; mean $\pm$ SD	5 $\pm$ 2	5 $\pm$ 1	1.06 (0.71-1.60)	.757		
Post-race urine color [‡] ; mean $\pm$ SD	4 $\pm$ 2	3 $\pm$ 2	1.31 (0.86-1.99)	.197		

Note: SD, standard deviation; RL, reference level; *statistically significant; ¥only among finishers; †specific gravity was categorized as 1 = 1.000, 2 = 1.005, 3 = 1.010, 4 = 1.015, 5 = 1.020, 6 = 1.025, 7 = 1.030; ‡urine color was categorized in an 8-scale category, 1 being light yellow and 8 being dark brown.

population. Baseline creatinine estimation is an important factor in determining whether or not the AKI criteria is met and eventually making appropriate clinical decisions. Further large-scale prospective studies are required to investigate these findings in different race environments with both pre- and post-race samples.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data available on request from the authors.

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